DSA Day 16: Recursion - fundamentals

Book: Data Structures Through C in Depth - Ch5 §5.1-5.4 · pp 147-162 | Code in Python

Overview

Recursion solves a problem via smaller instances of itself; needs a base case and a recursive case. The call stack stores frames.

Key concepts to master

- Base case & recursive case
- Call stack / frames
- Factorial, GCD, Fibonacci
- Towers of Hanoi

Python implementation

```
def hanoi(n,src,aux,dst):
    if n==0: return
    hanoi(n-1,src,dst,aux)
    print(src,'->',dst)
    hanoi(n-1,aux,src,dst)
```

Complexity

Hanoi O(2^n); factorial O(n).

Common pitfalls

- Missing/incorrect base case -> infinite recursion
- Deep recursion hitting the limit

Practice

- Trace Hanoi for 3 disks
- Draw the call tree of fib(5)